# Solving Recurrence Relations

There are 3 methods that we will learn to solve recurrence relations to find asymptotic time complexity:

1) Iteration method
2) Recursion tree
3) Master method.

## 1) Iteration Method:

We will using the example:

$$T(n) = \begin{cases} 0 & , n = 0 \\ c + T(n-1) & , n > 0 \end{cases}$$

for under standing.

There are in total 4 steps in this method to solve this.

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① Expand / iterate the recurrence relation 3 times atleast

$T(n) = c + T(n-1)$

$T(n-1) = c + T(n-2)$ 2nd iterate

$T(n-2) = c + T(n-3)$ 3rd iterate

$T(n-3) = c + T(n-4)$ 4th iteration

// we are only replacing n with (n-1), (n-2) & (n-3) in the equation.

② Express the iteration / expansion as a general formula in terms of k.

For this we can just start placing equation back into T(n).

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$T(n) = c + T(n-1)$

$= c + T(n-2)$

$T(n) = c + c + T(n-2)$

$= c + T(n-3)$

$T(n) = 3c + T(n-3)$

By looking at this, we should be able to infer the pattern.

$T(n) = kc + T(n-k)$.

general equation.

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③ Find "k" in terms of n. where problem reaches base case

base case:

$T(0) = 0$

$T(n-k) = T(0)$.

$n - k = 0$

$\boxed{n = k.}$

④ Substitute k back into the general formula & evaluate.

$= kc + T(n-k)$

substitute $k = n$

$= nc + T(n-n)$

$= nc + T(0)$, $T(0) = 0$ base case.

$= nc + 0$

$= nc$

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So our complexity: $\Theta(n)$

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Example 2:

$$T(n) = \begin{cases} 0, & n=0 \\ n + T(n-1), & n>0 \end{cases}$$

(1)

$T(n) = n + T(n-1)$

$T(n-1) = n-1 + T(n-2)$

$T(n-2) = n-2 + T(n-3)$

$T(n-3) = n-3 + T(n-4)$

(2)

$T(n) = n + T(n-1)$

$T(n) = n + (n-1) + T(n-2)$

$T(n) = n + (n-1) + (n-2) + T(n-3)$

$T(n) = n + (n-1) + (n-2) + (n-3) + T(n-4)$

$$T(n) = T(n-k) + \sum_{i=0}^{k-1} (n-i)$$ general equation

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(3) (base case)

$$T(n-k) = T(0)$$

$$n - k = 0$$

$$k = n.$$

(4)

$$T(n) = T(n-k) + \sum_{i=0}^{k-1} (n-i).$$

Substitute $k = n$.

$$T(n) = T(n-n) + \sum_{i=0}^{n-1} (n-i).$$

$$= 0 + \underbrace{[1+2+3 \cdots n]}_{\text{sum of arithmetic}}$$

$$= \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$$

$$\boxed{\Theta(n^2)}$$

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Example 3:

$$T(n) = \begin{cases} 1, & n=1 \\ 2T\left(\frac{n}{2}\right) + n, & n>1 \end{cases}$$

(1)

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \frac{n}{2}$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \frac{n}{4}$$

$$T\left(\frac{n}{8}\right) = 2T\left(\frac{n}{16}\right) + \frac{n}{8}$$

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(2)

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$T\left(\frac{n}{2}\right) = 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n = 4T\left(\frac{n}{4}\right) + 2n$$

$$T\left(\frac{n}{4}\right) = 2\left(2\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + \frac{n}{2}\right) + n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

$$\therefore \boxed{T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn.}$$

general equation.

(3)

$$T(1) = T\left(\frac{n}{2^k}\right) \quad \text{// if base can reached.}$$

$$\frac{n}{2^k} = 1$$

$$\boxed{k = \log_2(n).}$$

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(4)

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

// $k = \log_2(n)$

$$T(n) = 2^{\log_2(n)} T\left(\frac{n}{2^{\log_2(n)}}\right) + n\log_2(n)$$

$$T(n) = n + T(1) + n\log(n)$$

$$\boxed{\Theta(n \log n)}$$

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Example 4:

$$T(n) = \begin{cases} 1 & , n=1 \\ 2T\left(\frac{n}{2}\right)+1 & , n>1 \end{cases}$$

(1)

$$T(n) = 2T\left(\frac{n}{2}\right)+1$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right)+1$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right)+1$$

(2)

$$T(n) = 2\left(2T\left(\frac{n}{4}\right)+1\right)+1 = 4T\left(\frac{n}{4}\right)+3$$

$$T(n) = 2\left(2\left(2T\left(\frac{n}{8}\right)+1\right)+1\right)+1$$

$$= 8T\left(\frac{n}{8}\right) + 4+2+1$$

$$= 8T\left(\frac{n}{8}\right) + 7$$

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$= 2^k T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{k-1} 2^i$

remember sum of geometric series formula

$S_n = \frac{a(1-r^n)}{1-r}$

$2^k T\left(\frac{n}{2^k}\right) + \frac{1(1-2^k)}{1-2}$

$T(n) = 2^k T\left(\frac{n}{2^k}\right) + 2^k - 1$ — general formula.

(3)

$T(1) = 1$

$T(1) = T\left(\frac{n}{2^k}\right)$

$k = \log_2(n)$.

(4)

$T(n) = n^{\log_2(n)} T(1) + 2^{\log_2 n} - 1$

$= n + n - 1$

$= 2n - 1$

$\Theta(n)$

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Important formulas:

Sum of arithmetic series: (1, 2, 3, 4, 8)..

$$\frac{(a+b)n}{2}$$

Sum of geometric series (non-converging): (1, 2, 4, 8 ...)

$$\frac{a(1-r^n)}{1-r}$$

Sum of geometric series (converging). (1, 1/2, 1/4 ...).

$$\frac{a}{1-r}$$

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## 2) Recursion Tree Method

For this revise 3 formulas.

Total cost = Total cost of leaf nodes + cost of Total internal nodes.

↓

- cost of leaf nodes = cost of one leaf node * Total leaves.

cost of Total internal nodes = Sum of all levels.

so we will do this again with an example & it has — steps.

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Example 1:

$$T(n) = \begin{cases} 1, & n=1 \\ 2T\left(\frac{n}{2}\right) + cn, & n>1 \end{cases}$$

~~It shows create 2 recursive sub children.~~

① Expand term (like iterative method):

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \frac{cn}{2}$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \frac{cn}{4}$$

Also find $T(n)$, $T\left(\frac{n}{2}\right)$, $T\left(\frac{n}{4}\right)$

It shows general $T\left(\frac{n}{2^k}\right)$ so

$\frac{n}{2^k} = 1$

$k = \log_2 n$

② Find root & see number of childreps.

as we know $2T\left(\frac{n}{2}\right)$

This shows 2 recursive calls so every loads & to child, iteration each $T\left(\frac{n}{2}\right)$.

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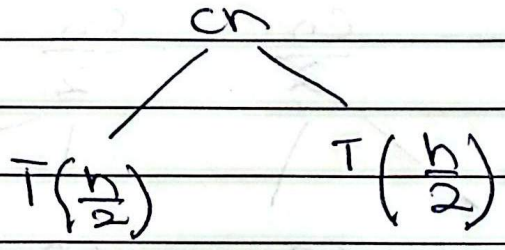
root: non recursive part of the equation:

root = c

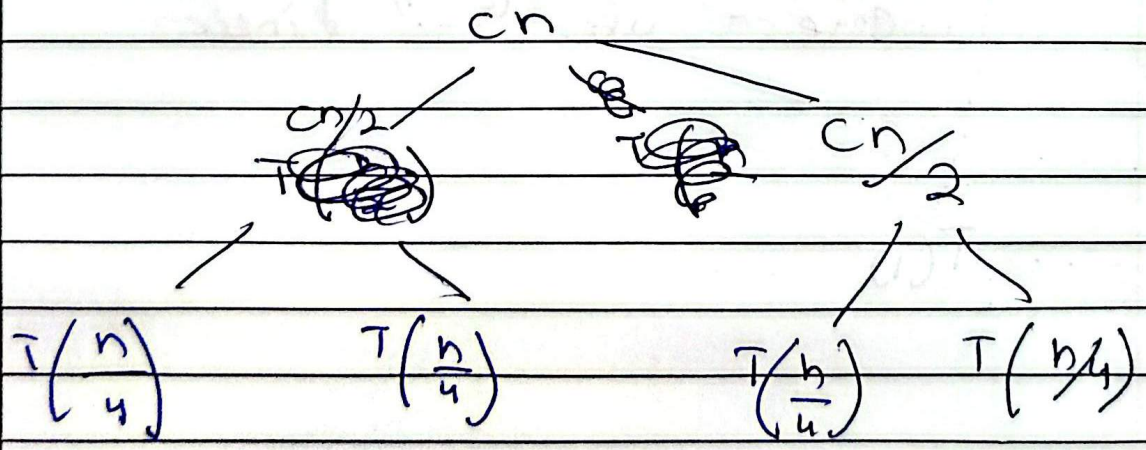



how keep on expanding.

$$T(n) = 2T\left(\frac{n}{4}\right) + \frac{cn}{2}$$

childs

new roots of sub trees

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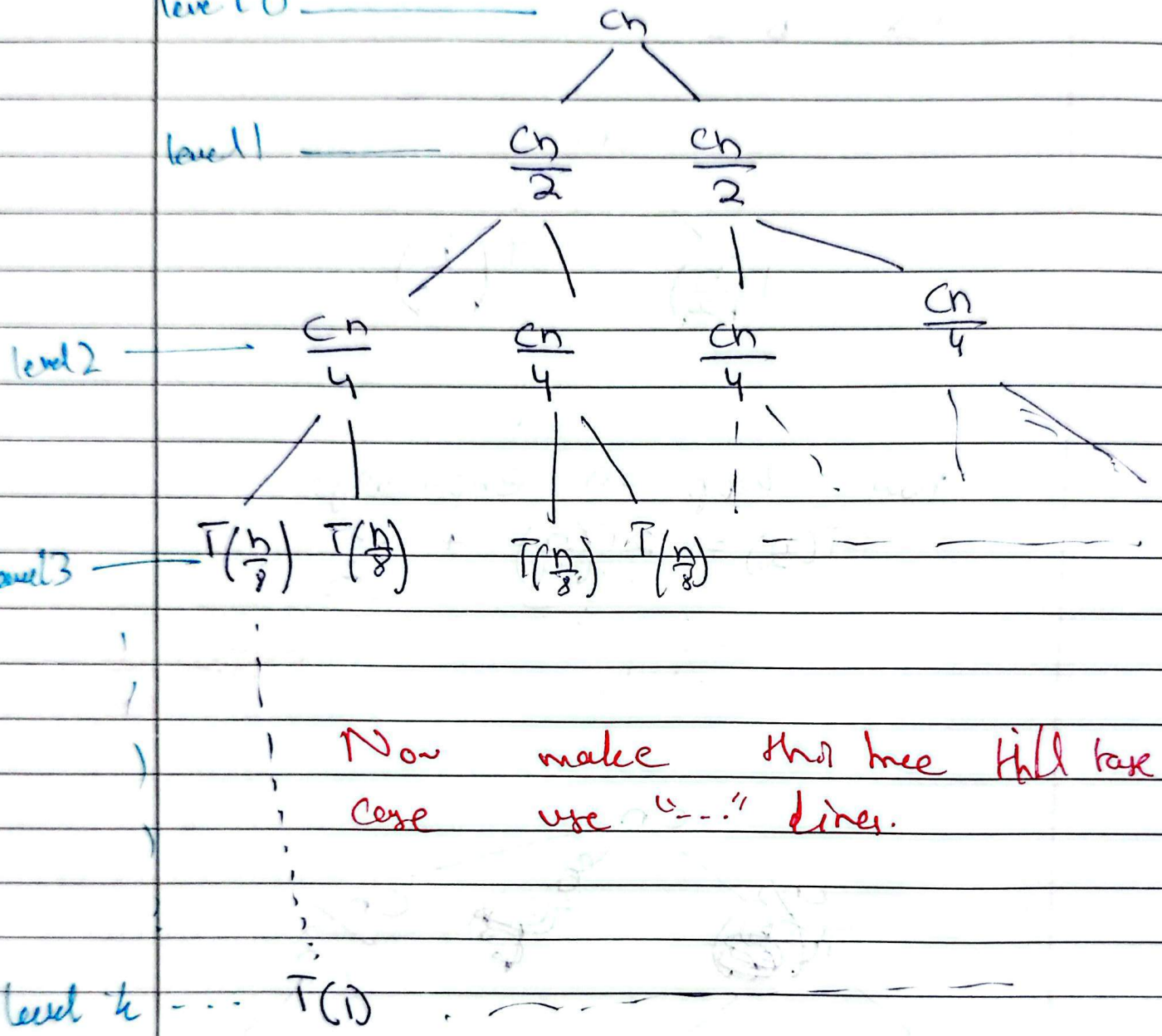
Repeat process again until 3 levels.




Now make this tree till last case use "..." dots.

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k is the total amount of iterations that we reach so last level is level k.

(3) Make a table

| levels | number of leaf nodes | cost |
|---|---|---|
| 0 | $2^0$ | cn |
| 1 | $2^1 = 2$ | cn/2 + cn/2 = cn |
| 2 | $2^2 = 4$ | cn/4 * 4 = cn |
| ⋮ | ⋮ | ⋮ |
| k | $2^k$ | cn |

cost of a single leaf node = 1

bcz base case. leaf node is base case

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④ Apply formula.

Total cost = cost of all leaves + cost of internal nodes.

$= 2^k * 1 + (cn + cn + cn \dots (k \text{ times}))$

$= 2^k + k\,cn.$

as $k = \log_2 n$

$= 2^{\log_2 n} + n c \log n$

$= n + nc \log n$

// ignore c

$= n + n \log n$

$\Theta(n \log n)$

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Example 2:

$$T(n) = \begin{cases} 1, & n = 1 \\ 2T\left(\frac{n}{2}\right) + n^2, & n > 1 \end{cases}$$

(1)

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \frac{n^2}{4}$$

$$T(n/4) = 2T\left(\frac{n}{8}\right) + \frac{n^2}{16}$$

Seeing $T(n), T\left(\frac{n}{2}\right), T\left(\frac{n}{4}\right)$

pattern is $T\left(\frac{n}{2^k}\right)$

$$\frac{n}{2^k} = 1$$

$$k = \log_2 n$$

In iterative method we used to derive a formula for the whole right side of equation. Here we only do for $T(n)$ to find $k$.

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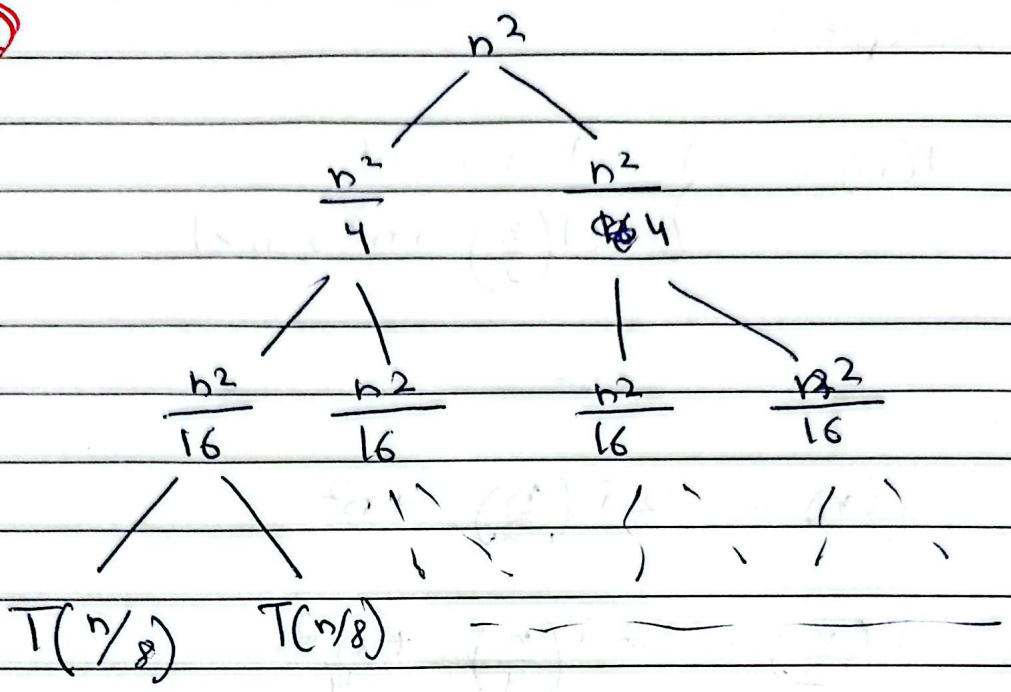
(2)




$T(1)$.

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(3)

| Levels | number of leaves | cost of leaves |
|---|---|---|
| 0 | $2^0 = 1$ | $n^2$ |
| 1 | $2^1 = 2$ | $\frac{n^2}{4} + \frac{n^2}{4} = n^2/2$ |
| 2 | $2^2 = 4$ | $n^2/4$ |
| . | . | . |
| . | . | . |
| . | . | . |
| k | $2^k$ | $n^2/2^k$ |

(4)

box (leaf cost).

Total cost = $2^k * 1 + n^2\left(1 + \frac{1}{2} + \frac{1}{4} \ldots\right)$

$= 2^k + n^2 \frac{1}{1 - 1/2}$ # sum of convergent geometric $\frac{a}{1-r}$

$= 2^{\log_2 n} + 2n^2$

$= n + 2n^2$

$\Theta(n^2)$.

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## Example 3:

$$T(n) = \begin{cases} 1, & n = 1 \\ 3T\left(\frac{n}{4}\right) + n^2, & n \geq 1 \end{cases}$$

so now 3 childs will be there

(1)

$$T(n) = 3T(n/4) + n^2$$

$$T(n/4) = 3 \cdot T\left(\frac{n}{16}\right) + \frac{n^2}{16}$$

$$T\left(\frac{n}{16}\right) = 3T\left(\frac{n}{64}\right) + \frac{n^2}{256}$$

### Finding k:

$T(n)$, $T(n/4)$, $T(n/16)$ ... $T(n/4^k)$.

$$T\left(\frac{n}{4^k}\right) = T(1)$$

$$k = \log_4 n.$$

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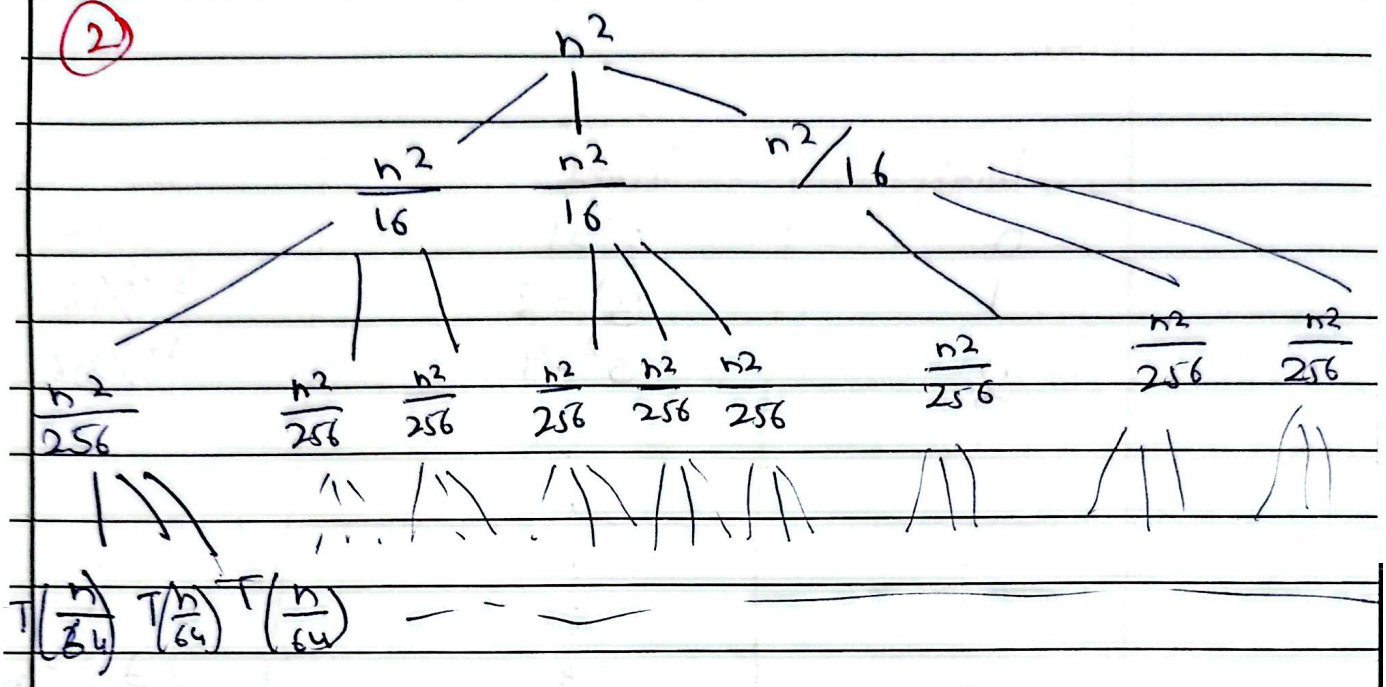
(2)


$n^2$
$\frac{n^2}{16}$
$\frac{n^2}{16}$
$n^2/16$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$\frac{n^2}{256}$
$T\left(\frac{n}{64}\right)$ $T\left(\frac{n}{64}\right)$ $T\left(\frac{n}{64}\right)$

$T(1)$

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(3)

| levels | leaves (count) | cost of level |
|---|---|---|
| 0 | $3^0=1$ | $n^2$ |
| 1 | $3^1=3$ | $3n^2/16$ |
| 2 | $3^2=9$ | $9n^2/256$ |
| ⋮ | ⋮ | ⋮ |
| k | $3^k$ | $\left(\frac{3}{16}\right)^k n^2$. |

you have to think this pattern out.

(4)

Total cost $= 3^k * 1 + n^2 \sum_{i=0}^{k} \left(3^i/16\right)^i$

even though k might be a small number we consider $k = \infty$, large number

$= 3^{\log_4 n} + n^2 \left(\frac{1}{1-\frac{3}{16}}\right)$ $\neq \frac{a}{1-r}$

$= 3^{\log_4 n} + n^2 \left(\frac{16}{13}\right)$

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$= 3^{\log_4(n)} + n^2\left(\frac{16}{13}\right)$

# property of log: $a^{\log_x(b)} = b^{\log_x(a)}$

$= n^{\log_4(3)} + \frac{16}{13}n^2$

$= n^{0.79} + \frac{16}{13}n^2$

$\Theta(n^2)$

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## Example 4: (Important)

$$T(n) = \begin{cases} 1, & n = 1 \\ T\left(\frac{n}{4}\right) + T\left(\frac{n}{2}\right) + n^2, & n > 1 \end{cases}$$

Now this is a very important example. In this both childs split unequally one is $T\left(\frac{n}{4}\right)$, while other is $T\left(\frac{n}{2}\right)$.

(1)

$$T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{n}{2}\right) + n^2$$

expand both seperately. as in tree they have seperate branches.

$$T\left(\frac{n}{2}\right) = T\left(\frac{n}{8}\right) + T\left(\frac{n}{4}\right) + \frac{n^2}{4}$$

$$T\left(\frac{n}{4}\right) = T\left(\frac{n}{16}\right) + T\left(\frac{n}{8}\right) + \frac{n^2}{16}$$

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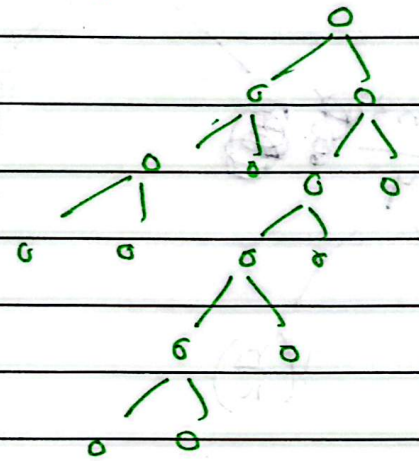
$$T\left(\frac{n}{8}\right) = T\left(\frac{n}{32}\right) + T\left(\frac{n}{16}\right) + \frac{n^2}{64}$$

(2) Tree:

Now tree will not end at a certain level. because of unequal splits.

might be like this:



so what will we consider k?

In Big-Oh, it will be biggest branch because we choose upper limit.

In Omega, it will be smallest terminating branch, because we finding lower limit.

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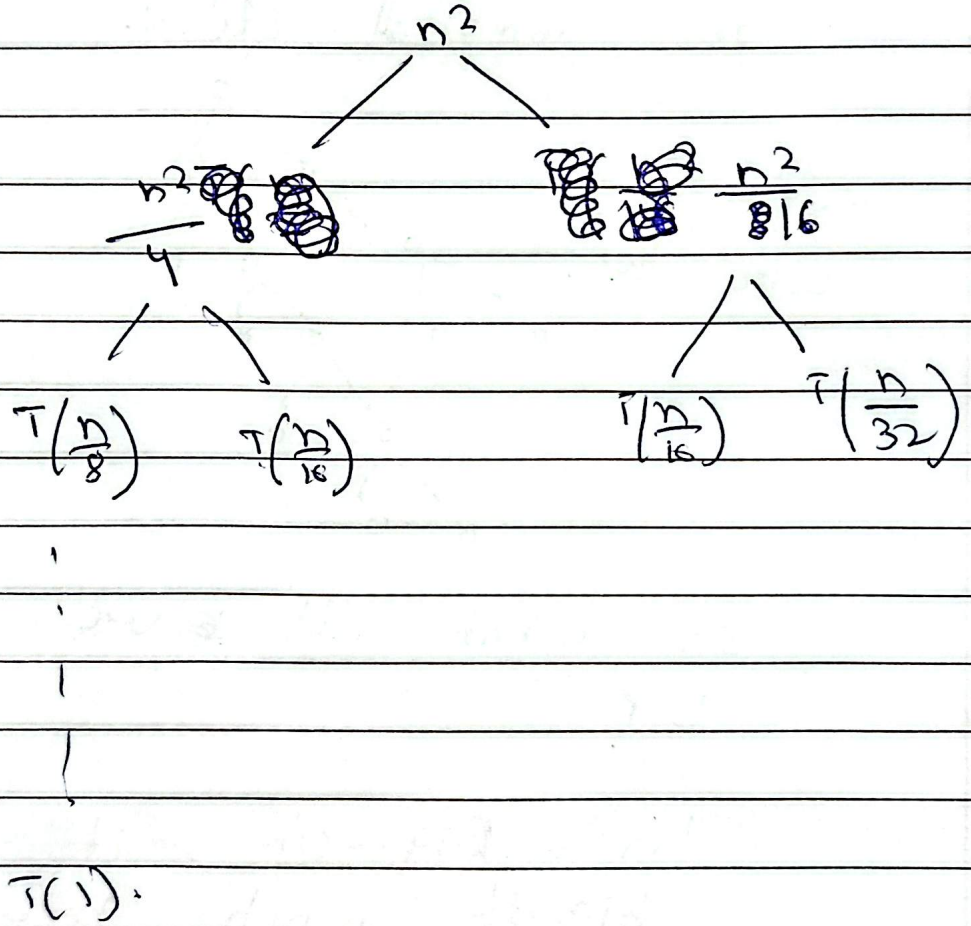
But how to know which will be smaller branch & which will be bigger?
we will decipher & infer that.

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Now from tree we can see that the branch:

$T(n) \rightarrow T(n/2) \rightarrow T(n/4) \rightarrow T(n/8)$... will be Biggest branch.

$T(n) \rightarrow T(n/4) \rightarrow T(n/16) \rightarrow T(n/64)$ --- will be Smallest branch.

now we will be calculating big-Oh so we take biggest branch & find k:

$$T\left(\frac{n}{2^k}\right) = T(1).$$

Pattern of biggest branch

$$k = \log_2(n).$$

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(3)

| Levels | Leaves | cost |
|---|---|---|
| 0 | $2^0 = 1$ | $n^2$ |
| 1 | $2^1 = 2$ | $n^2/4 + n^2/16 = 5n^2/16$ |
| 2 | $2^2 = 4$ | $25n^2/256$ |
| ⋮ | ⋮ | ⋮ |
| k | $2^4$ | $(5/16)^4 n^2$. |

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(4)

Total cost = $1 * 2^4 + n^2\left(1 + \frac{5}{16} + \frac{25}{256} \cdots\right)$

$= 2^{\log_2 n} + n^2\left(\frac{1}{1-\frac{5}{16}}\right)$

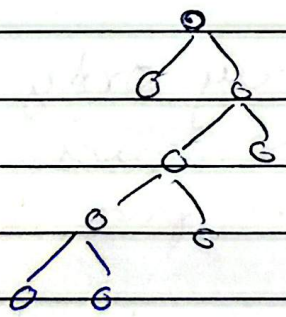
$= n + \frac{16}{11} n^2$

so

$O(n^2)$

OUR leaves calculation in this is WRONG!!!

Because a tree will be



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Our leaves will never be equal to $2^k$.

So how to calculate total leaves.

There are formulas for that too.

But for now revise one:

$$T\left(\frac{n}{2}\right) + T\left(\frac{n}{4}\right) + n^2$$

simply adds $\frac{n}{2}$ & $\frac{n}{4}$, $= \frac{3n}{4}$.

- This way only work if our recurse cases are in all fractions i.e in pure divide & conquer

* This is just a shortcut method.

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Example 5:

$$T(n) = \begin{cases} 1, & n <= 1 \\ T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n, & n > 1 \end{cases}$$

(1)

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n$$

$$T\left(\frac{n}{3}\right) = T\left(\frac{n}{9}\right) + T\left(\frac{2n}{9}\right) + \frac{n}{3}$$

$$T\left(\frac{2n}{3}\right) = T\left(\frac{2n}{9}\right) + T\left(\frac{4n}{9}\right) + \frac{2n}{3}$$

$$T\left(\frac{2n}{9}\right) = T\left(\frac{2n}{27}\right) + T\left(\frac{4n}{27}\right) + \frac{2n}{9}$$

$$T\left(\frac{n}{9}\right) = T\left(\frac{n}{27}\right) + T\left(\frac{2n}{27}\right) + \frac{n}{9}.$$

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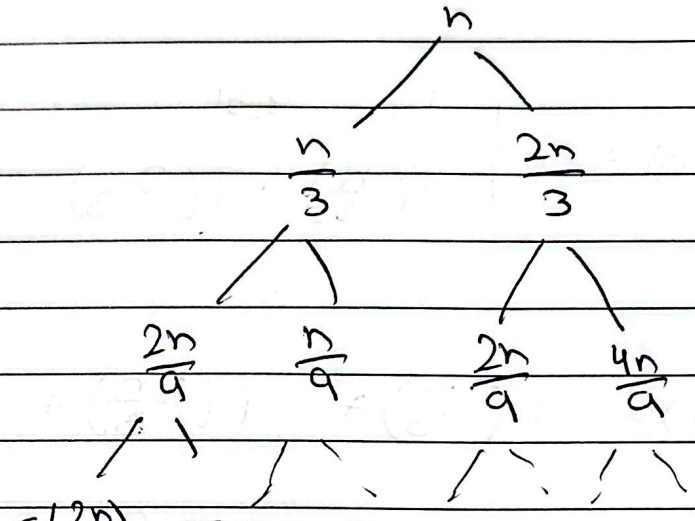
(2)




T(1)

now the largest branch will be

$$n \to \frac{2n}{3} \to \frac{4n}{9} \cdots$$

$$T(1) = T\left(\left(\frac{2}{3}\right)^k n\right)$$

$$\left(\frac{2}{3}\right)^k n = 1$$

$$\boxed{k = \log_{3/2}(n).}$$

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(3)

Now no need to make leaves colums bcz as all levels are not balanced.

Total leaves = $\left(\frac{n}{3}\right) + \frac{2n}{3} = n$

Total leaves cost = $n * 1 = n$

| levels | internal nodes cost |
|---|---|
| 0 | n |
| 1 | n |
| 2 | n |
| . | n |
| . | . |
| . | . |
| k | n |

Sum of all internal nodes = $kn$

so Total cost = $n + n\log_{3/2}(n)$

so $\theta(n \log n)$.

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IMP:

If we would have taken all leaves ending at max branch it would have led to:

$2^k$ leaves.

&

$2^{\log_{3/2}(n)} = n^{1-71}$

& $n^{1-71} > n \log_{3/2}(n)$

which is wrong answer as & we will never reach that complexity.

Because our leaves are only $\Theta(n)$ not $n^{1-71}$.

So in & unequal splits remember that mostly our complexity is only due to cost of Internal nodes & leaves cost ignored or you can learn how to calculate total leaves & find them out too.

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## 3) Master Theorem:

This is the easiest ~~form~~ way to solve divide & conquer recurrency of form,

$$T(n) = aT\left(\frac{n}{b}\right) + f(n).$$

$a >= 1$
$b > 1$
} This must be true & a, b are constants always.

The master theorem basically directly compares the 2 costs:

$f(n)$: (The root cost).

$n^{\log_b(a)}$: ((leaf cost) (Total work done in base case leaf).

$X = n^{\log_b a}$ = leaf cost

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Now from this comparison 3 cases come:

① $f(n) = O\left(n^{\log_b(a) - \varepsilon}\right)$ for some $\varepsilon > 0$

What we are saying here is that $f(n)$ grows polynomially slower than $n^{\log_b(a)}$ by a factor on $n^{\varepsilon}$.

Its answer is always,

$$T(n) = \Theta\left(n^{\log_b(a)}\right).$$

we mean in this that if we $\left(\frac{n^{\log_b(a)}}{n^{\varepsilon}}\right)$ divide our leaf cost by a factor $n^{\varepsilon}$ then its growth rate will be equal to $f(n)$.

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Thus this shows our leaf cost is bigger than the root cost, hence by rule of summations in complexity analysis:

$$\Theta(n^{\log_b(a)}).$$

(2) $f(n) == \Theta(n^{\log_b(a)})$ =>

This means a balanced tree, both are growing at the same rate.

In this case answer is:

$$\Theta(n^{\log_b(a)} \log^{k+1} n), \quad k >= 0$$

now k over here is the power of log in f(n)

eg

$$T(n) = 2T\left(\frac{n}{3}\right) + \log^2 n$$

$k = 2$.

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③ $f(n) = \Theta\left(n^{\log_b(a)+\varepsilon}\right)$ , $\varepsilon > 0$

In this case our f(n) / Root cost is grow polynomially faster than our leaf cost.

In this we can see that our $f(n) = n^{\log_b(a)} * n^{\varepsilon}$ which show f(n) is bigger by $n^{\varepsilon}$ times.

In this answer.

$$\Theta(f(n))$$

In this last 3rd case we also need to verify:

$$a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n).$$

for some $c < 1$ & large n.

a & b values are from the question. find a value of c that satisfies it & proved.

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All 3 are well defined formulas by 3 Mathematicians in 1980.

In summary:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$a \geq 1$
$b > 1$
} always check condition first or else Master theorem fails.

$$X = n^{\log_b(a)}$$

(1) if $f(n) < X$,

$$\Theta(n^{\log_b(a)})$$

(2) if $f(n) \equiv X$

$$\Theta(n^{\log_b(a)} \log^{k+1} n)$$

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(3) if $f(n) > x$

$\Theta(f(n))$.

& verify:

a. $f\left(\frac{n}{b}\right) \leq c \cdot f(n)$.

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Example 1:

1) $4T\left(\frac{n}{2}\right) + n$

$X = n^{\log_2 4} = n^2$

as $n^2 > n$

so case 1 ( $f(n) < X$ )

Hence

$\boxed{\Theta(n^2)}$

2) $4T\left(\frac{n}{2}\right) + n^3$

$X = n^{\log_2 4} = n^2$ , $b = 2, a = 4$

as $n^3 > n^2$

so case 3.

$\boxed{\Theta(n^3)}$

but as it is case 3 we need to verify.

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a. $f\left(\frac{n}{b}\right) \leq c \cdot f(n)$

$4 \cdot \left(\frac{n}{2}\right)^3 \leq c \cdot n^3$

$\frac{n^3}{2} \leq c \cdot n^3$

so $c \geq \frac{1}{2}$

e.g even take $c = \frac{1}{2}$ & verify.

$\frac{n^3}{2} \leq \frac{n^3}{2}$

$n^3 \leq n3$ ✓

True so proved

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(3) $4T\left(\frac{n}{2}\right) + n^2$

$f(n) = n^2$ , $a = 4$, $b = 2$

$X = n^{\log_b a} = n^2$

Case 2:

as $f(n)$ has no log so power of log is 0

$k = 0$

$\Theta(n^2 \log n)$

(4) $T\left(\frac{n}{2}\right) + 2^n$

$f(n) = 2^n$ , $a = 1$, $b = 2$

$X = n^{\log_2 1} = n^0 = 1$

as $f(n) > X$ so

case 3.

$\Theta(2^n)$.

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verify:

$a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n)$

$1 \cdot 2^{n/2} \leq c \cdot 2^{n}$

$1 \cdot \frac{2^n}{4} \leq c \cdot 2^n$

$c \geq \frac{1}{4}$

take $c = \frac{1}{4}$

$\frac{2^n}{4} \leq \frac{2^n}{4}$

$2^n \leq 2^n$ ✓

True so verified.

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③ $2^n T\left(\frac{n}{2}\right) + n$

$a = 2^n$, a should be a constant

So Master Theorem can't be applied.

⑥ $\sqrt{2}\, T\left(\frac{n}{2}\right) + \log n$

$a = \sqrt{2}, \quad b = 2$

$X = n^{\log_2 \sqrt{2}} = \sqrt{n}$

as $\sqrt{n} > \log n$

so Case 1.

$\Theta(\sqrt{n})$

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Very Imp.

(7) $T(n) = 2T\left(\frac{n}{2}\right) + n\log n$

$a = 2$, $b = 2$, $f(n) = n\log n$

$X = n^{\log_b a} = n$.

As we know

$n\log n > n$

So case 3:

but $f(n)$ should be polynomially $\log n$ than $n^{\log_b a}$. i.e there should be a $\varepsilon$ that satisfies.

$f(n) = n^{\log_b (a+\varepsilon)}$

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so our

$$\varepsilon = \frac{n}{n \log n} = \frac{1}{\log n}$$

but this will always lead to $\Theta$ bcoz we take $n$ as $\infty$ & $\log(\infty) = \infty$ & $\frac{1}{\infty} = 0$.

but our $\varepsilon > 0$ so case 3 fails here.

as $\varepsilon$ is so small. so we consider this as case 2.

$$\Theta = n \log^2(n)$$

$k = 1$. so $n \log^{k+1}(n)$

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(8) $2T\left(\frac{n}{2}\right) + \frac{n}{\log n}$

$X = n^{\log_2 2} = n.$

$f(n) = \frac{n}{\log n}$

Now as $n > \frac{n}{\log n}$, case 1 should.

apply but :

$f(n) = \Theta(n^{\log_b (b-\epsilon)})$

should be such's fixed.

ratio = $\frac{n}{\log n} = \log n$

now our

$n^{\epsilon} \leq \log n$

should be true.

But for all large n's & $\epsilon > 0$, this will fail

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Here case 1 doesn't apply.

case 2 will also fail because

$$\frac{n}{\log n} = n \log^{-1} n.$$

$k = -1$

but our $k$ should be $\geq 0$.

Hence Master Theorem can't solve this.

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Here case 1 doesn't apply

Case 2 will also fail because

$$\frac{n}{\log n} = n \log^{-1} n.$$

$k = -1$

but our $k$ should be $\geq 0$.

Hence Master Theorem can't solve this.

day/date 3/7

- This call is happening beau $\frac{n}{\log n}$ is logarithmically slower than n.

But we need Polynomilly slower.

If it was

$f(n) = \frac{n}{\log n}$ or $n \log n$

& $x = n^2$

now $n^2$ is polynomially larger than $\frac{n}{\log n}$ or $n \log n$

hence we apply Master theorem.

Conclude: Polynimally larger is required! If loguithimically larger than we cannot apply master theorem.

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(9) $2T\left(\frac{n}{2}\right) + \frac{n^2}{\log n}$

$X = n^{\log_2 2} = n^1$

now $\frac{n^2}{\log n}$ & $n$ are polynomially different with $f(n) > X$

so

$$\Theta\left(\frac{n^2}{\log n}\right)$$

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Imp.

(10) $2T\left(\frac{n}{2}\right)$ $-n^2$

problem

$x = n^1$

eventhough our $|f(n)|$ is bigger than x the master theorem requires our $f(n)$ to be

Positive.

If negative sign comes, master theorem fails.

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